

Enact Framework

*A Project Report submitted in partial fulfilment of the requirements
for the award of the degree of*

Bachelor of Technology in Computer Science and Engineering by

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Declaration

I hereby declare that the work which is being presented in the Internship is in partial fulfilment of the requirements for the award of the Bachelor of Technology in Computer Science and Engineering and submitted to the Department of Computer Engineering and Applications of GLA University, Mathura, is an authentic record of my own work carried under the supervision of **Ms. Athulya P(Team Lead, LG Soft India).**

The contents of this project report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree.

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ACKNOWLEDGEMENT

The **internship** opportunity I have with **LG Soft India** is a great chance for learning and professional development. Therefore, I consider myself as a very lucky individual as I was provided with an opportunity to be a part of it. I am also grateful for having a chance to meet so many wonderful people and professionals who led me through this internship period.

Bearing in mind previous I am using this opportunity to express my deepest gratitude and special thanks to the **Lead, Athulya P** who in spite of being extraordinarily busy with his duties, took time out to hear, guide and keep me on the correct path and allow me to carry out my project at their esteemed organization.

I perceive this opportunity as a big milestone in my career development. I will strive to use gained skills and knowledge in the best possible way, and I will continue to work on their improvement, in order to attain desired career objectives.

Sincerely,

Nikhil

Intern

LG Soft India

ABSTRACT

Smart TV platforms have grown into highly sophisticated ecosystems that demand seamless performance, intuitive user interfaces, and tight system-level integration. **webOS**, developed by **LG Electronics**, powers millions of smart televisions globally, making its application ecosystem a critical component of the overall user experience.

During this internship, I worked on system-level applications within the webOS platform, contributing to **feature enhancements, bug fixes, UI improvements, and testing workflows**. The development stack primarily included **Enact** (a React-based framework tailored for TV applications), along with **JavaScript**, **CSS**, and **JSX**. I also utilized webOS Developer Tools for debugging and deployment, and Jest for writing and maintaining unit tests.

My contributions focused on enhancing UI components to deliver a smoother and more responsive user experience across different screen sizes and resolutions. I identified and resolved critical issues affecting **application stability** and **performance**, ensuring consistent behavior across multiple LG TV models. Additionally, I implemented unit and automation test cases, which helped improve code reliability and reduced regression issues during releases.

This experience gave me hands-on exposure to real-world application development in an embedded smart TV environment. It also strengthened my understanding of large-scale front-end architecture, performance optimization, and the unique challenges of building applications for non-traditional platforms like televisions, where factors such as remote-based navigation, limited hardware resources, and varied display configurations play a significant role.

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CHAPTER -1: INTRODUCTION

1.1 Motivation

Working at **LG Soft India** exposed me to the domain of **Smart TV application development**, a field I was previously unfamiliar with. Understanding how system applications work in a TV environment, how remote-based navigation differs from touch interfaces, and how performance matters on low-resource devices provided a refreshing and stimulating challenge.

The opportunity to work on **Enact-based LG TV applications** motivated me to learn new technologies, understand TV-oriented UI/UX principles, and contribute meaningfully to production-grade features. Every new task—whether feature enhancement, debugging, UI development, or writing test cases—brought practical learning and encouraged continuous improvement.

1.2 Overview

The MS System App Solutions Team at LG Soft India is responsible for developing and maintaining applications that run on LG webOS Smart TVs. These applications provide essential system-level functionalities and ensure a seamless user experience.

The major focus areas include:

- Building and maintaining applications using Enact (React-based framework)
- Ensuring compatibility across different LG TV models and screen resolutions
- Enhancing UI components and system features
- Fixing bugs and performance bottlenecks
- Writing unit and automation test cases to ensure reliability

1.3 About the Organization

LG Soft India (LGSI) is the official software research and development center of LG Electronics, located in Bangalore, India. As a key contributor to LG's global innovation ecosystem, LGSI develops software solutions for a wide range of LG products, including smart devices, consumer electronics, AI-enabled systems, and home entertainment technologies. The center is committed to delivering high-quality, reliable, and user-centric software that supports LG's vision of technological excellence.

Within LGSI, the **Media Solutions (MS) Department** focuses on the development of high-performance applications for **LG Smart TVs**, ensuring optimal functionality, seamless user experience, and compatibility across diverse global models. The department works extensively on the webOS platform, using modern JavaScript frameworks, UI technologies, and device-level optimization techniques.

I have been working as an intern in the **MS System App Solutions Team**, where I contributed to the enhancement and maintenance of LG TV system applications. My role involved working with the **Enact framework (built on React)**, developing UI components, resolving issues, improving features, and writing unit and automation test cases. This experience provided valuable exposure to industry-grade development practices and the operational workflow of large-scale consumer software products.

CHAPTER - 2: PROJECTS UNDERTAKEN

2.1. Remote Control Guide Application

Technologies: Enact, JavaScript, React, CSS Modules, webOS APIs

Overview:

The Remote Control Guide application is designed to help users understand the functionality of TV remote buttons. When a user presses any button on the remote, the application dynamically displays a contextual description on the screen, improving usability and accessibility.

Role & Contributions:

- Enhanced navigation flows specifically optimized for remote-based interactions
- Implemented real-time UI feedback for remote key presses
- Improved focus management and directional navigation handling

Outcome: Delivered a responsive and user-friendly guidance system that improved the overall remote usability experience for end users.

2.2. Photobooth Application

Technologies: Enact, JavaScript, React, CSS Modules, webOS APIs

Overview:

The Photobooth application allows users to capture photos using the TV, apply different layouts and frames, and send the final images to a Photoism store for printing.

Role & Contributions:

- Developed and enhanced UI components for photo capture and preview flows
- Implemented layout and frame selection features for customizable outputs
- Ensured seamless navigation using remote control inputs
- Optimized UI rendering for large screen displays
- Contributed to improving performance and responsiveness during image processing flows

Outcome: Delivered a responsive and user-friendly guidance system that improved the overall remote usability experience for end users.

2.3. Bug Fixing & Issue Resolution

Technologies: JavaScript, JSX, Enact, Chrome DevTools, webOS IDE

Key Tasks:

- Fixed layout responsiveness issues across multiple TV resolutions
- Resolved component lifecycle and rendering inconsistencies
- Eliminated remote-navigation issues (focus handling, directional key behavior)

Outcome:

Improved overall application stability and reduced user-facing defects across both applications.

2.4. UI Component Development

Technologies: Enact UI Components, React Hooks, CSS Modules

Key Tasks:

- Built reusable UI components tailored for the TV environment
- Implemented screen-friendly designs compatible with large displays
- Ensured accessibility for remote-based navigation across all components

Outcome: Created maintainable, scalable components used across different modules.

2.5. Unit Testing for TV Applications

Technologies: Jest, Enzyme/React Testing Library

Work Done:

- Wrote unit test cases for UI components and utilities
- Achieved improved test coverage for critical modules
- Ensured functionality correctness through component-level testing

Outcome: Increased reliability and reduced chances of regression issues.

CHAPTER 3 : SOFTWARE REQUIREMENT ANALYSIS

3.1 Technical Feasibility

Frontend Technologies

- Enact Framework (based on React)
- React.js, JavaScript (ES6+)
- HTML5, CSS3, JSX

Backend / Platform

- LG webOS TV Platform APIs
- Node.js for build tools

Testing

- Jest
- Automation Testing Tools for webOS

Tools & Environment

- webOS IDE
- VS Code
- Git

2.2 Software Used

Minimum recommended system requirements:

- Intel Core i5 processor
- 8 GB RAM
- Windows 10 / Ubuntu 20.04
- Chrome Browser
- VS Code
- webOS TV Emulator
- Git version control

CHAPTER 4 : PROJECT DESIGN

LG Smart TV applications run on the **webOS platform**, which uses a web-technology-based architecture. The applications developed in the **MS System App Solutions Team** follow a highly modular, component-oriented approach using the **Enact framework**. The project design is structured to support scalability, performance, and smooth remote-navigation experiences required for Smart TV devices.

While internal architecture diagrams and proprietary workflows cannot be disclosed due to LG confidentiality policies, this chapter presents a generalized and safe-to-share design overview that reflects the actual development approach used during the internship.

4.1 Overview of Smart TV Application Architecture

LG Smart TV applications built on webOS use Enact, a React-based framework optimized for TV environments.

Unlike traditional web applications, Smart TV apps must address unique constraints such as remote-based navigation, strict performance requirements, varying screen resolutions, localization, and limited hardware resources.

To efficiently handle these challenges, the application is designed using a layered and modular architecture that ensures scalability, responsiveness, and maintainability.

4.2 Component-Based Architecture

The TV application follows a **component-driven architecture**, where each UI screen and functionality is broken into modular, reusable components.

Major component categories:

1. Presentational Components

- Responsible only for displaying UI
- No business logic
- Utilize Enact's Moonstone UI library

- Styled using CSS Modules

2. Container Components

- Manage application state
- Handle event interactions
- Pass data to presentational components
- Trigger navigation and system actions

3. Utility & Helper Modules

- Reusable JavaScript utility functions
- Focus management helpers
- API wrappers
- Validation logic
- Common formatting functions

4. Layout Components

- Screen-level structures
- Arrangement of UI blocks
- Responsive scaling for different TV resolutions

Benefits of this architecture:

- High code reusability
- Simplified debugging
- Easy unit testing
- Consistent UI behavior across modules

4.3 Focus & Navigation Design (TV Remote Interaction)

Remote-based navigation is a core part of every Smart TV application. Enact provides **Spotlight**, a navigation engine that manages focus across components.

Key Design Elements:

- Registers each focusable element within the system
- Handles directional navigation (Up, Down, Left, Right)
- Maintains default focus states
- Controls focus transitions between screens
- Prevents focus traps and dead zones

Why this is important:

Since TVs do not support touch or mouse input, smooth and predictable focus navigation is essential for usability, accessibility, and overall user experience.

4.4 Screen & Routing Design

The application uses an organized routing structure to manage navigation between screens.

Screen Design Principles:

- Each screen is implemented as an independent Enact/React component
- Lightweight components are loaded first, followed by heavier modules
- Navigation is controlled using routing utilities and history-based transitions

Routing Types Used:

- Static routing for core system screens
- Dynamic routing for user-driven flows
- Conditional routing based on device state or configurations

This ensures fast screen transitions and reduces runtime overhead.

4.5 State Management Design

Smart TV applications require efficient state handling to maintain responsiveness.

State Management Approach Used:

- Local state management using React Hooks

- Context API for sharing global data
- Memoization using useMemo and useCallback to avoid unnecessary re-renders
- Derived state optimization for performance

What kind of state is managed:

- UI selection and active focus
 - API data (system details, configurations, etc.)
 - Local temporary data
 - User interactions and preferences
-

4.6 API Interaction & Data Flow

Although the TV apps are mostly frontend-based, certain modules fetch or update data using **webOS system APIs**.

API Communication Design:

- API requests are wrapped inside reusable services
- Promise-based async/await flow
- Error handling at a centralized location
- Loading indicators for slow operations
- Fallback UI in case of API failure

Data Flow Cycle:

- Component triggers an API call
- Service layer communicates with webOS API
- Response is processed and validated
- Cleaned data is passed to UI components
- UI updates based on new state

This ensures consistent and predictable behavior across the app.

4.7 UI/UX Design Considerations for TV

Designing for Smart TVs is fundamentally different from mobiles and desktops.

Key UI Requirements:

- Large screen layouts optimized for distance viewing
- Minimal text density
- High-contrast color usage
- Intuitive component placement
- Clear focus indicators
- Fixed-position menus for easy navigation

LG webOS-specific UI considerations:

- Using Moonstone UI library for consistent native look
- Ensuring components are accessible with remote
- Avoiding deeply nested UI structures
- Maintaining 60 FPS rendering wherever possible

4.8 Performance Optimization Design

Smart TVs have more limited resources than PCs or phones, so performance optimization is crucial.

Performance Techniques Used:

- Lazy loading of heavy components
- Avoiding unnecessary re-renders (React.memo, useCallback)
- Reducing DOM tree depth
- Preloading assets for critical screens
- Efficient list rendering using virtualized lists
- Minimizing CSS complexity

Outcome:

- Reduced rendering delays
- Faster navigation transitions

- Improved overall user experience
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4.9 Testing Design

Testing was integrated into the design stage to ensure reliability.

Unit Testing Design:

- Component-level snapshot testing
- Event and user-action testing
- Mocking API calls for isolation
- Testing spotlight navigation logic

Automation Testing Design:

- Script-based testing for repetitive navigation flows
- Validation across different resolutions and models
- Regression verification for modified screens

This ensured stability and minimized bugs in production deployments.

4.10 Deployment & Build Design

The build and packaging process is handled using the Enact CLI and webOS tools.

Build Process:

- Code is compiled using Babel
- Transpiled and optimized JavaScript is generated
- CSS modules are bundled
- Output is packaged into IPK (webOS application format)

Deployment Targets:

- LG webOS Emulator
- Real LG Smart TV devices

Ensuring compatibility across multiple hardware generations is a key part of the deployment process.

4.11 Summary of Project Design

The entire project follows:

- Modular, component-driven development
- Optimized navigation for remote control
- Efficient state and rendering management
- Secure and abstracted API interaction
- Rigorous unit and automation testing
- Performance-focused architecture

This combination enables LG TV applications to be fast, intuitive, visually appealing, and scalable across millions of devices globally.

CHAPTER 5 : PROJECT TECH STACK

The LG TV application ecosystem is built on top of **LG webOS**, which uses web technologies to deliver high-performance system applications for Smart TVs. During the internship, multiple modern tools, frameworks, and libraries were used for development, testing, debugging, and application lifecycle management. The following sections describe each major technology in detail.

5.1 Frontend Technologies:

Enact Framework

Enact is LG Electronics' official UI framework built **on top of React** and optimized specifically for **webOS Smart TV applications**. It provides a structured architecture and a rich set of UI components tailored for TV remote control interfaces.

Key Features of Enact:

- **SandStone UI Library:** A specialized component library optimized for TV screens, large displays, and remote navigation.
- **Spotlight System:** Handles directional navigation (Up, Down, Left, Right) using the remote. This is essential for TV apps where pointer or touch interaction is limited.
- **High Performance Rendering:** Enact optimizes React rendering to run smoothly even on limited hardware inside TVs.
- **Internationalization (i18n):** Built-in utilities support localization and translation for global TV models.
- **Reusable Components:** Encourages modular development with reusable building blocks.

Why Enact?

- Designed specifically for the TV environment
 - Smooth navigation and consistent user experience
 - Reduces development complexities of smart TV UI
-

React.js

React is a JavaScript library for creating interactive UI components. Enact extends React, so understanding React's fundamentals is essential.

Key Concepts Used:

- **Component-Based Architecture:** Applications are composed of reusable functional components.
- **Virtual DOM:** Improves update efficiency by minimizing DOM manipulation.
- **Hooks (useState, useEffect, useCallback, etc.):** Used to manage state, lifecycle events, and memoization.
- **Props & State Management:** Ensures predictable behavior across components.

Applications in LG TV Development:

- Creating TV UI screens
 - Handling focus and state transitions
 - Developing modular and maintainable components
-

JavaScript (ES6+)

JavaScript powers the entire logic layer of webOS applications. ES6+ features significantly enhanced development productivity.

Major JavaScript Features Utilized:

- Arrow functions
- let/const variable declarations
- Spread/rest operators
- Promises & async/await
- Array and object destructuring
- Modular imports/exports

Usage in the Project:

- Writing component logic
 - Handling async TV system APIs
 - Managing UI interactions and event flows
 - Implementing validation and utility functions
-

HTML5

HTML5 provides the structural foundation of the application.

Key Capabilities Used:

- Semantic markup
- Audio/video element support
- Enhanced rendering performance
- Offline and cache-related features

HTML5 forms the underlying foundation of the Enact-React TV application.

CSS3

CSS3 is used to create visually appealing and responsive TV interfaces.

Major CSS Concepts Utilized:

- Flexbox Layout (heavily used in TV UIs)
- Grid Layout
- CSS Modules (scope-based styling)
- Animations and transitions
- Theming and color management

Enact-Specific

Styling:

Enact uses **CSS Modules**, where each component has its own stylesheet scoped locally to avoid conflicts.

5.3 Testing Tools: Unit Testing & Automation Testing

Jest (Unit Testing)

Jest is a JavaScript testing framework used to validate UI components, utility functions, and interaction logic.

Key Features Used:

- Snapshot testing for components
- Mocking functions and API calls
- Testing event handlers and focus behaviors
- Ensuring correct component rendering

Impact:

Helped ensure stability and prevent regressions during new feature development.

Automation Testing Framework for webOS

Automation testing is crucial for validating:

- Remote navigation
- Focus behavior
- Performance
- Feature functionality across multiple TV models

Key Areas Covered:

- Directional key flows (Up/Down/Left/Right navigation)
- Component focus consistency
- Cross-resolution screen compatibility
- Behavior under different system configurations

Outcome:

Significantly reduced manual testing workload and improved reliability.

5.4 Development Tools & Utilities

webOS IDE / Emulator

Used for:

- Running LG TV apps on virtual devices
- Testing multiple resolutions
- Debugging TV navigation and UI behavior

VS Code

Primary development editor

- Extensions used: ESLint, Prettier, React Snippets, GitLens

Git

Version control for:

- Branching strategy
- Code Reviews
- Collaboration with team members

Git tools: Git Bash, Git GUI, Gerrit

5.5 Summary of Tech Stack

The project leverages a modern, scalable, and performance-focused technology stack consisting of:

- Enact and React for UI development
- JavaScript for application logic
- HTML5 and CSS3 for structure and styling
- Jest and automation frameworks for testing
- webOS tools and Git for development and deployment

This combination ensures the delivery of high-quality, efficient, and scalable Smart TV applications optimized for millions of users worldwide.

CHAPTER 6 : CONCLUSION & PROJECT OUTCOME

6.1 Conclusion

During the course of this internship, I had the opportunity to work on real-world system applications built on webOS at LG Soft India. This experience allowed me to actively contribute to production-level features while gaining practical exposure to large-scale Smart TV application development.

Working with Enact significantly strengthened my understanding of React-based architectures and component-driven design. I developed a deeper grasp of building scalable UI systems, handling remote-based navigation, and optimizing applications for constrained hardware environments such as Smart TVs.

In addition to development, I improved my debugging and problem-solving skills by identifying and resolving complex UI and functional issues. I also gained hands-on experience in writing unit and automation test cases using Jest, which enhanced my understanding of test-driven development and software reliability.

Overall, the internship provided a well-rounded learning experience, combining technical growth with exposure to industry practices, collaborative workflows, and production-grade development standards.

6.2 Project Outcome

The work carried out during this internship resulted in measurable improvements to LG Smart TV applications and contributed to their overall quality and performance.

- Successfully implemented feature enhancements across multiple system applications, improving usability and functionality
- Identified and resolved UI and functional bugs, leading to a more stable and seamless user experience
- Designed and developed reusable, modular UI components that can be leveraged across different applications
- Gained practical expertise in Enact framework, webOS APIs, and Smart TV-specific UI/UX design principles

This experience not only strengthened my technical foundation but also equipped me with the skills required to work effectively on large-scale, performance-critical frontend systems.

CHAPTER-7 : BIBLIOGRAPHY

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